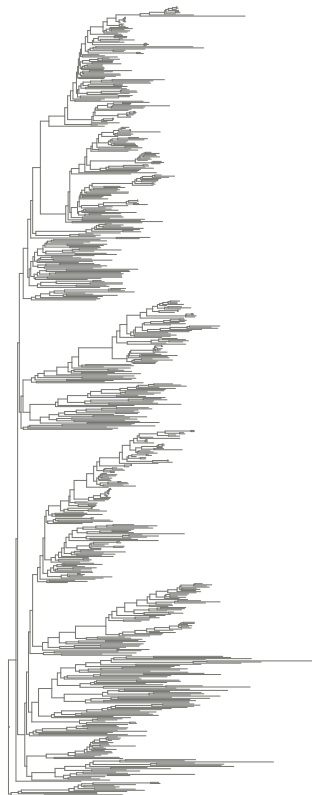


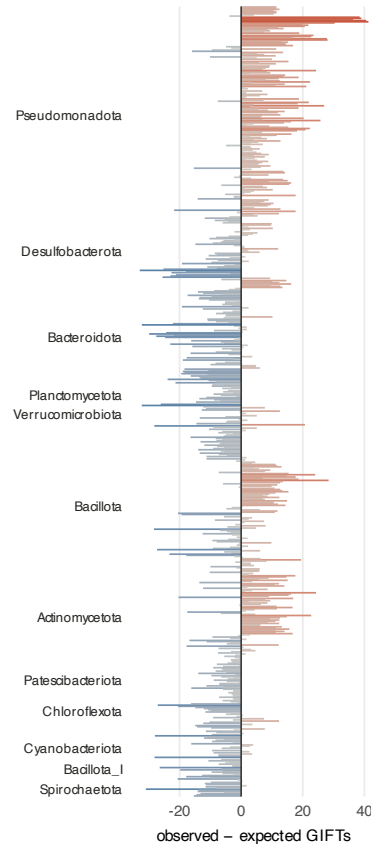
Deviation from the size expectation is phylogenetically clustered

Standardised deviation of 696 genomes from the Figure S9 expectation: Pagel's $\lambda = 0.94$ (likelihood-ratio $P < 0.001$), Blomberg's $K = 0.714$ (permutation $P < 0.001$). The band in c is the 95% range of 9,999 tip permutations.

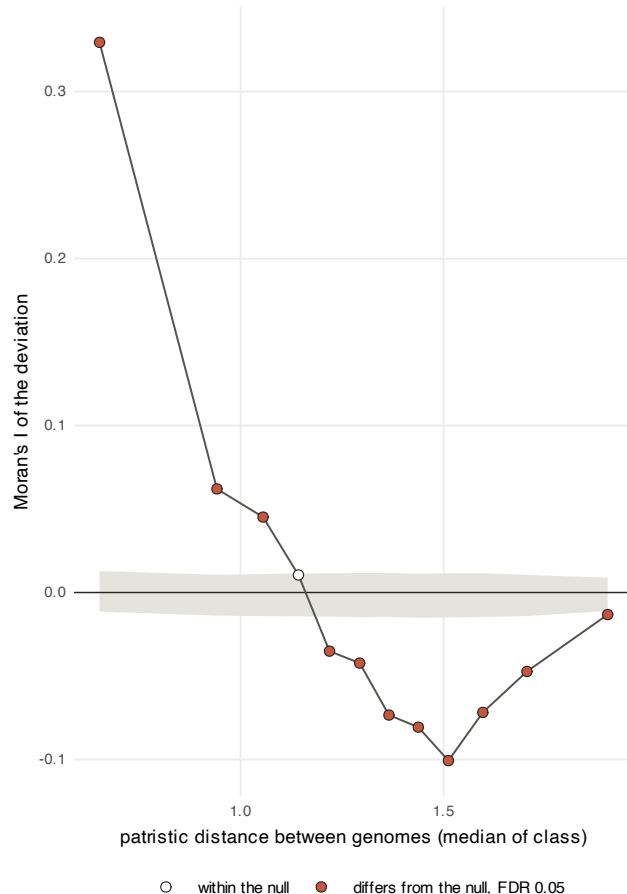
a GTDB bac120 tree



b Deviation



c Depth of the clustering



Positive Moran's I means that genomes at that distance deviate in the same direction. Clustering describes how encoded, curated capabilities are distributed over this panel; it is not an ancestral-state reconstruction.